
Individual Report — Bharath Gandhimani(35501308)

FIT5225 Cloud Computing and Security — Assignment 2

Role and Contribution

My primary responsibility was the complete AWS backend infrastructure. This included designing and deploying all nine Lambda functions — deduplication, inference, thumbnail, upload, query, tags, notifications, presign, and delete — along with the S3 bucket architecture, DynamoDB schema, SNS topic with species-based filter policies, API Gateway endpoints, Cognito user pool, and the ECR container image pipeline built on EC2.

The most significant technical challenge I faced was the thumbnail Lambda. It had never successfully executed — first due to a missing Pillow dependency, then because building the deployment package on a Mac M-series produced an aarch64 binary that Lambda's x86_64 runtime rejected. The fix required rebuilding the zip inside Docker using the official AWS Lambda Python 3.12 image with the `--platform linux/amd64` flag. A secondary issue arose from AWS Academy resetting credentials every four hours, which caused AccessDenied errors that initially looked like permission failures but were actually caused by missing files and stale DynamoDB records. Addressing this required a session-start protocol to reapply the bucket policy and flush orphaned records.

The other major architectural fix was replacing a public S3 bucket with a fully private bucket accessed through a dedicated presign Lambda. This eliminated both the security exposure and the session-reset problem in a single change.

Teamwork Reflection

The team functioned well throughout the project. Communication was consistent — we held regular standup calls and used GitHub to keep everyone aligned on progress. Each member took ownership of a defined area, which kept work from overlapping and made dependencies easier to manage.

My closest collaboration was with Gokhul, who built the React frontend. We worked continuously together to ensure the API contracts between the backend and frontend were correct, particularly around the presigned URL flow and query response formats. Prem handled the GCP Cloud Function proxy and also supported Gokhul with UI testing once the backend was stable.

Melissa and I collaborated closely on DynamoDB, SNS, and Cognito. A pivotal contribution from her was suggesting the dedicated presign Lambda as the solution to the public bucket problem. It was the right call — it resolved the security issue, eliminated the session-reset overhead, and required no DynamoDB migration since raw S3 paths were already being stored. Her input on the architecture diagram was also valuable, and she led its construction with the flow I provided.

The S3 bucket access issue and the repeated Docker rebuilds for the inference Lambda were the most frustrating points in the project. Both consumed more time than they should have. In hindsight, I would have invested more time upfront planning the Lambda dependencies, runtime environments, and access architecture before writing any code. Brute-forcing fixes after the fact cost days that better planning could have saved. That said, the team held together well under pressure. Everyone contributed, no one disengaged, and we delivered a working system.